Revision Notes for Class 11 Biology

Chapter 1 – The Living World

Life is a unique process that is made from the aggregation of molecules. These molecules undergo various chemical reactions to perform their specific functions which are called metabolism. This results in the production and utilization of energy. The metabolism will result in the growth, development, reproduction, adaptations, etc of living organisms through the production of various biomolecules.

All the living organisms that live in various habitats are found to share a somewhat similar genetic material that may be either terrestrial, or aquatic, in mountains, deserts, oceans, forests, etc.

Living organisms contain certain important characteristics that include growth and development, body organization, homeostasis, reproduction, adaptation, and energy utilization.

Diversity in the Living World:

The Earth is the main area where living organisms live. The world consists of millions of living organisms that sometimes we cannot even see with our naked eyes. These organisms are found to be living in various habitats that include forests, oceans, deserts, lakes, mountains, and even hot water springs.

There are different types of plants, animals, and insects in the world. This is very important and their variability is necessary for survival. The number of species identified and studied is 1.7-1.8 million. They all together make a natural diversity of life in the world which is generally called biological diversity or biodiversity.

Taxonomy: Taxonomy is the study of the classification, characterization, nomenclature, and identification of organisms and it is a branch of science. Systematics is another branch of science that includes the study of the classification, nomenclature, identification, and evolutionary history of an organism. Thus, the taxonomic characteristics of an organism along with its evolutionary history come under systematics. In 1813, A.P de Candolle was the first to introduce the term taxonomy while systematics was introduced at the time of human civilization.

The term Systematics is derived from the Latin word 'systema' which means the systematic arrangement of organisms. Linnaeus published his book Systema Naturae where the classification of plants, and animals was based on taxonomy.

Neo-systematics is the branch of systematics that deals with the species to be the product of evolution. In 1940, Julia Huxley was the one who developed this concept. It involves the known characteristics of an organism and also the known evidence from different fields of biology.

Identification: It is the method of pacing the organisms in their exact place based on their classification. The identification of organisms can be done with the help of taxonomic keys.

Classification: The classification is the process of grouping various living organisms based on the common features that they share. A single group consists of those organisms that have similar common features. To make classification easier various groups are formed in which different organisms are placed depending upon their characteristics.

Characterization: The studying and understanding of characters of organisms and categorizing them like external and internal structure (morphology and anatomy), the structure of the cell (cytology), developmental process (embryology), and ecological information (ecology) of the organism.

Nomenclature (naming): The naming of living organisms is called nomenclature. There are two types of names, one is vernacular (common names) and the other is a scientific name. Local names are used in local languages or common language and are easy for the local peoples but these names are not used by biologists because:

1. For many species a single local name is often used.

2. The local names sometimes lead to incorrect meanings about the organism.

3. In different regions of the country or world, different local names are used for one organism. Scientific names: The names are given according to certain rules and are followed by the biologist all over the world. To make it common around the world various international codes have been established.

These codes are:

• ICBN-International Code of Botanical Nomenclature

• ICZN-International Code of Zoological Nomenclature

• ICVN-International Code of Viral Nomenclature

• ICBN/ICNB-International Code for Bacteriological Nomenclature or Nomenclature of Bacteria.

Binomial Nomenclature:

Carl Linnaeus is credited for the introduction of the binomial nomenclature of plants and animals with his work in the book Species Plantarum in 1753. Binomial nomenclature is the biological system of naming organisms in which the name is composed of two terms, where, the first term indicates the genus and the second term indicates the species of the organism. E.g., Mangifera indica Linn. Mangifera is the genus name and indica is the species name. Linn indicates that this species was first described by Linnaeus Who can give scientific

names: Identification, nomenclature, and classification of organisms are all involved in this branch of biology.

Rules:

1. A scientific name generally has two components (words) in Latin or is derived from Latin irrespective of their origin.

2. The First word of the biological name denotes the genus name whereas the second one denotes species.

3. When applying the binomial nomenclature system, the name of the species is written in italics or underlined separately when handwritten.

4. The generic name must start with a capital letter while a specific name should start with a small letter.

5. The name of the author is printed in Roman or an abbreviated form at the end after the species name.

6. Only one correct name must be assigned to each taxonomic group.

7. The scientific name selected should be such that it would be easy to pronounce, and short.

Eg: Mangifera indica- Mangifera is the genus name and indica is the species name.

Taxonomic Categories:

In 1956 the term Taxon was introduced and in 1964, Mayr defined taxon to be the various categories based on different characters of the organisms that consist of a taxonomic group of any rank.

Taxonomic Hierarchy:

Various organisms in different categories depend upon their common characters to make classification easier. These groups together are called taxonomic hierarchies. The taxonomic hierarchy includes. Kingdom, division of the kingdom, phylum, class, order, family, genus, and species. Species are the lowest while the kingdom is the highest rank within the hierarchy. It is also called the Linnaean hierarchy as it was first proposed by Carolus Linnaeus, the Father of Systematic Botany. The hierarchy includes seven obligate categories. They are as follows-

Kingdom - Animalia

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Phylum - Chordata

(Division in case of plants)

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Class - Mammalia

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Order - Primata

↑

Family - Hominidae

↑

Genus - Homo

↑

Species – sapiens

- **Species**: It is the lowest category of the taxonomic hierarchy. There are around 8.7 million species observed on earth till now while the rest are left undiscovered. It refers to a group of organisms that are similar in shape, form, generative options. Species may be divided into subspecies. It was first defined by Ernst Mayr in 1964 that the species are

the interbreeding populations that are reproductively isolated from other such groups. The term species was first introduced by the biologist John Ray.

E. g.: sapiens.

- **Genus**: A category that is placed above species as they consist of a group of related species. Genus are of various types based on the number of species present like monotypic (one genus present), and polytypic (several species present). For e.g., the genus Panthera constitutes both lion and tiger.
- **Family**: This taxonomic category consists of related genera having similar characteristics. For e.g., the families Canidae, Felidae, Ursidae, etc come under one order Carnivora.
- **Order or Cohort:** This taxonomic category is more specific than the class as it consists of one or more similar families. The class Mammalia consists of around twenty-six orders that include primates, Carnivora, etc.
- **Class**: It was the most general taxonomic category before the introduction of phyla. In the animal kingdom, there are around 108 classes that include Pisces, reptilia, aves, etc. The categories used in classification now are different from those of the Linnaeus taxonomy.
- **Phylum**: This category is more specific than the kingdom. In the animal kingdom, there are around thirty-five phyla that include phylum Arthropoda, Chordata, etc.
- **Kingdom**: The highest level of classification is the kingdom which is further divided into various subgroups. The total kingdoms of the living organisms are five in number including Monera, Protista, Fungi, Plantae, and Animalia.

Species - *Homo sapiens*

Genus - Homo

Family - Hominids

Order - Primates

Class - Mammals

Phylum - Chordates

Kingdom - Animals

Generic Name	Specific Epithet	Common Name
Mangifera	indica	Mango

Solanum	tuberosum	Potato
Solanum	nigrum	Nightshade
Panthera	leo	Lion
Panthera	tigris	Tiger
Homo	sapiens	Man

Common Name	Biological Name	Genus	Family	Order	Class	Phylum/ Division
Man	Homo sapiens	Homo	Hominidae	Primata	Mammalia	Chordata
Housefly	Musca domestica	Musca	Muscidae	Diptera	Insecta	Arthropoda
Mango	Mangifera indica	Mangifera	Anacardiaceae	Sapindales	Dicotyledonae	Angiospermae
Wheat	Triticum aestivum	Triticum	Poaceae	Poales	Monocotyledonae	Angiospermae